

EduGenie Project

Topic 4 — quiz_module.py — Quiz Generation

```
import json
from pydantic import BaseModel, Field
from gemini_client import get_client
from config import GEMINI_MODEL
from google.genai import types

class QuizQuestion(BaseModel):
    question: str
    options: list[str] = Field(min_length=4, max_length=4)
    correct_answer: str
    explanation: str

class QuizResponse(BaseModel):
    questions: list[QuizQuestion] = Field(min_length=3, max_length=3)

def clean_json_block(text: str) -> str:
    text = text.strip()
    if text.startswith("```"):
        lines = text.splitlines()
        if lines and lines[0].startswith("```"):
            lines = lines[1:]
        if lines and lines[-1].strip() == "```":
            lines = lines[:-1]
        text = "\n".join(lines).strip()
    return text

def generate_quiz(content: str) -> dict:
    prompt = f"""
Create exactly 3 multiple-choice questions for students from the following topic or passage.
Each question must have exactly 4 options.
The correct_answer must exactly match one option.
Add a short explanation of the correct answer.
Keep questions clear and educational.

Content:
{content}
"""
    client = get_client()
    response = client.models.generate_content(
        model=GEMINI_MODEL,
        contents=prompt,
        config=types.GenerateContentConfig(
            temperature=0.4,
            response_mime_type="application/json",
            response_schema=QuizResponse,
        ),
    )

    if getattr(response, "parsed", None) is not None:
        quiz = response.parsed
        return quiz.model_dump() if hasattr(quiz, "model_dump") else quiz

    raw = clean_json_block(response.text or "")
    try:
        data = json.loads(raw)
        quiz = QuizResponse.model_validate(data)
        return quiz.model_dump()
    except Exception as exc:
        raise RuntimeError(f"Could not parse Gemini quiz JSON: {exc}")
```